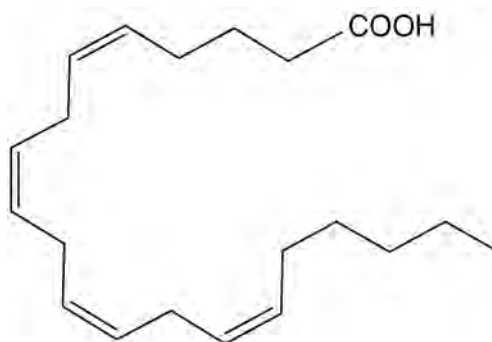


0 3 . 1 **Figure 2** shows a fatty acid that contains 20 carbon atoms and four double bonds.

Figure 2

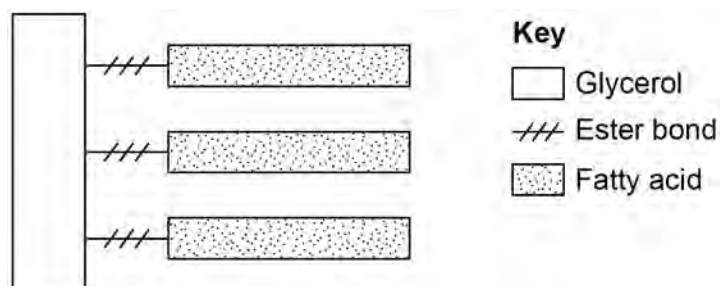


On **Figure 2**, draw a box around the R group of the fatty acid.

[1 mark]

0 3 . 2 **Figure 3** shows a triglyceride.

Figure 3



Describe **two** differences between the structure of the triglyceride shown in **Figure 3** and a phospholipid.

[2 marks]

1 _____

2 _____

Question 3 continues on the next page

Turn over ►



0 3 . 3

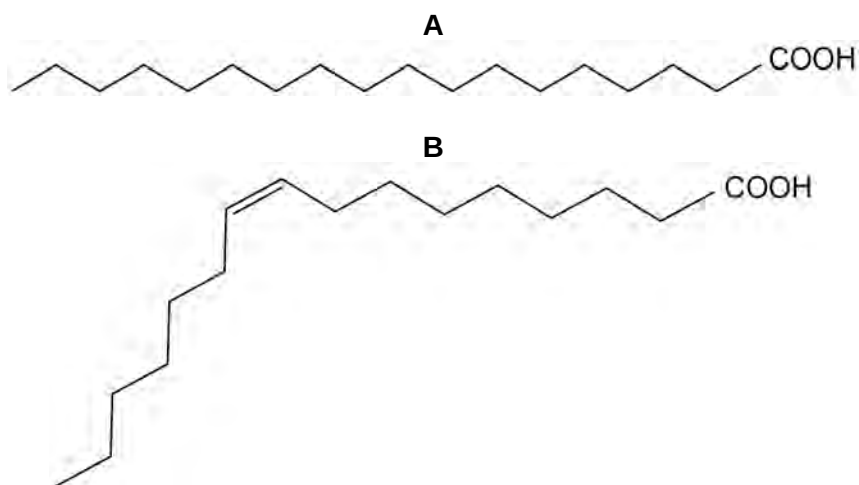
Explain why phospholipids can form a bilayer but triglycerides cannot.

[3 marks]

0 3 . 4

Figure 4 shows two fatty acids, **A** and **B**.

Figure 4



Scientists fed rats a diet with added fish oil for 4 months.

They obtained samples of red blood cells from the rats before starting this diet (0 months) and after 4 months on this diet.

For each red blood cell sample, they separated the cell-surface membranes and measured:

- the percentage of phospholipids containing each of the fatty acids **A** and **B**
- the fluidity of the membrane.



